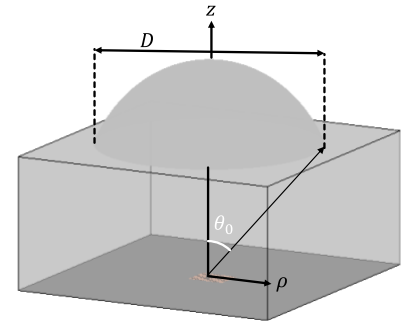


## Assignment 3: Lens Antenna

Write a MATLAB code to analyze an antenna placed at the lower focal point of an elliptical lens made of a material with permittivity of  $\epsilon_r$ . Suppose that this antenna feed is operating at 200 GHz and its far field inside an infinite medium, with permittivity of  $\epsilon_r$ , has the following expression:



$$\vec{E}_{far}^a(\theta, \phi) = e^{-\left\{\left[\frac{u}{u_0}\right]^2 + \left[\frac{v}{v_0}\right]^2\right\}} (\sin\phi\hat{\theta} + \cos\phi\hat{\phi}) \frac{e^{-jk_ar}}{r}$$

where  $k_d = \sqrt{\epsilon_r}k_0$ ,  $u = \sin\theta\cos\phi$  and  $v = \sin\theta\sin\phi$  and  $e^{-\left\{\left[\frac{u}{u_0}\right]^2 + \left[\frac{v}{v_0}\right]^2\right\}}$  represents a Gaussian shaped far field, with  $u_0, v_0$  parameters modeling its directivity in the two main planes ( $\phi = 0^\circ, \phi = 90^\circ$ ).

## Provide the following results in the report:

1. The far field of the planar antenna in fused silica ( $\epsilon_r = 4$ ) infinite medium for  $u_0 = v_0 = 0.5$ .
2. Compute the Fresnel transmission coefficients for a flat fused silica-air interface. Plot the power transmitted over the power incident as a function of the incident plane wave angle for both TE and TM polarized cases. Identify and explain the points of interest in this figure.
3. Compute the Fresnel transmission coefficients for an untruncated elliptical fused silica lens with diameter of  $D = 9\lambda_0$ . Plot the power transmitted over the power incident as a function of the lens elevation angle for both TE and TM polarized cases. Compare these results with the ones of Question 2.
4. Plot the equivalent aperture current distribution of an elliptical lens antenna with  $\epsilon_r = 4$ ,  $D = 9\lambda_0$ ,  $\theta_0 = 40^\circ$ , and the described feed with  $u_0 = v_0 = 0.5$ , in the main planes of the antenna. Describe how the distribution of these currents is affected by the feed's far field (Question 1) and the transmission coefficients of the lens.
5. Plot the far field patterns radiated by the lens antenna of Question 4. Compare it with the far field of an equivalent uniform aperture (airy pattern). Describe what aspects govern the (possible) asymmetry in this far field pattern.
6. Give an estimation of the directivity and gain of the lens antenna of Question 4.

*Assignment's report should contain your comments on the obtained results. Providing only the requested plots does not count as a complete report.*